

JEE MAIN REVISION MASTER: LINEAR PROGRAMMING

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Objective Function, Constraints & Feasible Region	14 PYQs	Core Principle
Moderate	Corner Point Theorem & Optimal Solutions	5 PYQs	Optimization Archetypes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Objective Function, Constraints & Feasible Region	$Z = ax + by$, $\text{Feasible Region: Intersection of all half-planes determined by constraints}$
Corner Point Theorem & Optimal Solutions	$\text{If } Z(A) = Z(B) = Z_{\max}, \text{ then } Z \text{ is maximized at all points on segment } AB$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Objective Function, Constraints & Feasible Region (High Yield — 14 PYQs)

Core Principle: Formulation of LPP: linear objective function $Z = ax + by$, linear constraints, non-negative restrictions $x, y \geq 0$, and identification of bounded vs unbounded feasible regions.

■ **Caution / JEE Pitfall:** If the feasible region is unbounded, the minimum or maximum value of Z may not exist unless the open half-plane has no common points with the feasible region.

■ **Speed Trick:** In KCET, sketch the constraint boundary lines quickly using intercept form $(x/a) + (y/b) = 1$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points ..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "The corner points of the feasible region determined by the system of linear constraints are $(0,10),(5,5),(15,15),(0,20)$. Let $z=px+qy$ wh..." **[Ans: D]**
- **[2025 Shift] Q#3 (MCQ):** "Consider the following statements: Statement (I): In a LPP, the objective function is always linear. Statement (II): In a LPP, the linear ine..." **[Ans: N/A]**

3.2 Corner Point Theorem & Optimal Solutions (Moderate Yield — 5 PYQs)

Core Principle: Corner Point Method: If an optimal value exists, it occurs at a vertex. If the maximum/minimum occurs at two distinct corner points, it attains the same optimal value at every point along the line segment joining them.

■ **Caution / JEE Pitfall:** Assuming optimal solution is only at integer points—fractional coordinates of corner points are valid.

■ **Speed Trick:** When $Z = px + qy$ attains the same maximum at (x_1, y_1) and (x_2, y_2) , equate $p x_1 + q y_1 = p x_2 + q y_2$ to find p/q directly.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points ..." [Ans: N/A]
- **[2024 Shift] Q#1 (MCQ):** "Corner points of the feasible region for an LPP are $(0,2), (3,0), (6,0), (6,8)$ and $(0,5)$. Let $Z=4x+6y$ be the objective function. The min..." [Ans: N/A]
- **[2022 Shift] Q#3 (MCQ):** "The corner points of the feasible region of an LPP are $(0,2), (3,0), (6,0), (6,8)$ and $(0,5)$, then the minimum value of $z=4x+6y$ occurs at..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points at which $Z_{\text{(max)}}$ occurs is

- (A) 1
- (B) 2
- (C) 0
- (D) Infinity

Correct Answer: None

Solution Analysis: If the objective function $Z=ax+by$ has the same maximum value at two corner points of the feasible region, then Z will have the same value at every point on the line segment joining those two corner points. This happens because $Z=ax+by$ is a linear function. So, if it is maximum at two endpoints of an edge of the feasible region, then it remains maximum at all points on that edge. Hence, $Z_{\text{(max)}}$ occurs at infinitely many points. So, the correct option is: Option D: Infinity

Benchmark Problem 2: KCET 2020 (Q#3)

The feasible region of an LPP is shown in the figure. If $z=11x+7y$, then the maximum value of Z occurs at

- (A) (0, 5)
- (B) (3, 3)
- (C) (5, 0)
- (D) (3, 2)

Correct Answer: None

Solution Analysis: Given, maximize $z=11x+7y$ Intersecting point of lines $x+y=5$ and $x+3y=9$ is $(3,2)$. $\therefore$ Corner point is $B(3,2)$ For corner points of the feasible region We put, $x=0$ in $x+3y=9$ $y=3 \Rightarrow$ corner point is $A(0,3)$ and put $x=0$ in $x+y=5$, we get $y=5 \Rightarrow$ corner point is $C(0,5)$ The values of z at these corner points are as follows Corner points $z=11x+7y$ $A(0,3)$ $11 \times 0 + 7 \times 3 = 21$ $B(3,2)$ $11 \times 3 + 7 \times 2 = 47$ $C(0,5)$ $11 \times 0 + 7 \times 5 = 35$ Therefore, maximum value of z is 47 at $(3, 2)$.